



ECEN 314 – Signals and Systems

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Module 16 - Application: Analog Filter Design
Part 2 – More Butterworth LPFs + Chebyshev LPFs

Learning Objectives



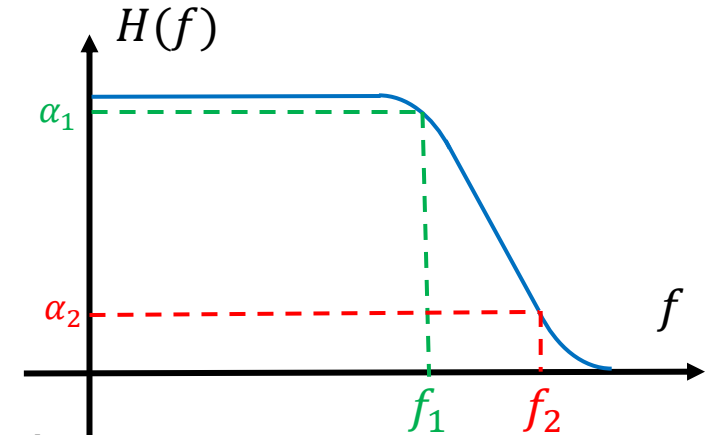
- Given specifications on the passband and the stop band, be able to design a Butterworth LPF to meet those specifications.
- Understand the form and characteristics of a Chebyshev low-pass magnitude response.
- Given specifications on the passband and the stop band, be able to design a Chebyshev LPF to meet those specifications.

LPF Design Using Butterworth Filters



Suppose we wish to design a LPF using Butterworth Filters to meet a generic set of design specs of the form:

- Passband: $|f| < f_1$, $|H(f)|^2 > \alpha_1$,
- Stopband: $|f| > f_2$, $|H(f)|^2 < \alpha_2$.



For the Butterworth Filter, the magnitude response is

$$|H(f)|^2 = \frac{1}{1 + \left(\frac{f}{f_3}\right)^{2n}}.$$

The first step is to determine what the order of the filter should be. If we just met the above specs (with equality) we would have

$$|H(f_1)|^2 = \frac{1}{1 + \left(\frac{f_1}{f_3}\right)^{2n}} \geq \alpha_1 \rightarrow \left(\frac{f_1}{f_3}\right)^{2n} \leq \frac{1}{\alpha_1} - 1 \rightarrow f_3 \geq \frac{f_1}{\left(\frac{1}{\alpha_1} - 1\right)^{1/2n}}$$

$$|H(f_2)|^2 = \frac{1}{1 + \left(\frac{f_2}{f_3}\right)^{2n}} \leq \alpha_2 \rightarrow \left(\frac{f_2}{f_3}\right)^{2n} \geq \frac{1}{\alpha_2} - 1 \rightarrow f_3 \leq \frac{f_2}{\left(\frac{1}{\alpha_2} - 1\right)^{1/2n}}$$

LPF Design Using Butterworth Filters



In order to satisfy the design specifications, the 3dB frequency must be chosen to satisfy:

$$\frac{f_1}{\left(\frac{1}{\alpha_1} - 1\right)^{1/2n}} \leq f_3 \leq \frac{f_2}{\left(\frac{1}{\alpha_2} - 1\right)^{1/2n}}$$

This will only be possible if the lower bound is smaller than the upper bound. Therefore, it is also required that:

$$\frac{f_1}{\left(\frac{1}{\alpha_1} - 1\right)^{1/2n}} \leq \frac{f_2}{\left(\frac{1}{\alpha_2} - 1\right)^{1/2n}}$$

To ensure that this is satisfied, the order of the filter is chosen according to

$$n \geq \frac{1}{2} \log \left(\frac{\frac{1}{\alpha_2} - 1}{\frac{1}{\alpha_1} - 1} \right) / \log \left(\frac{f_2}{f_1} \right)$$

LPF Design Using Butterworth Filters



The minimum required value of n given by the formula on the previous slide will generally not be an integer, so you will need to round up to the next highest integer. Using $\lceil \cdot \rceil$ to denote the ceiling (round up) function, the minimum order of the Butterworth filter needed to meet the given specifications $\{f_1, \alpha_1, f_2, \alpha_2\}$ is

$$n = \left\lceil \frac{1}{2} \log \left(\frac{\frac{1}{\alpha_2} - 1}{\frac{1}{\alpha_1} - 1} \right) / \log \left(\frac{f_2}{f_1} \right) \right\rceil$$

Once n is chosen, the 3dB frequency is then chosen to satisfy

$$\frac{f_1}{\left(\frac{1}{\alpha_1} - 1 \right)^{1/2n}} < f_3 < \frac{f_2}{\left(\frac{1}{\alpha_2} - 1 \right)^{1/2n}}$$

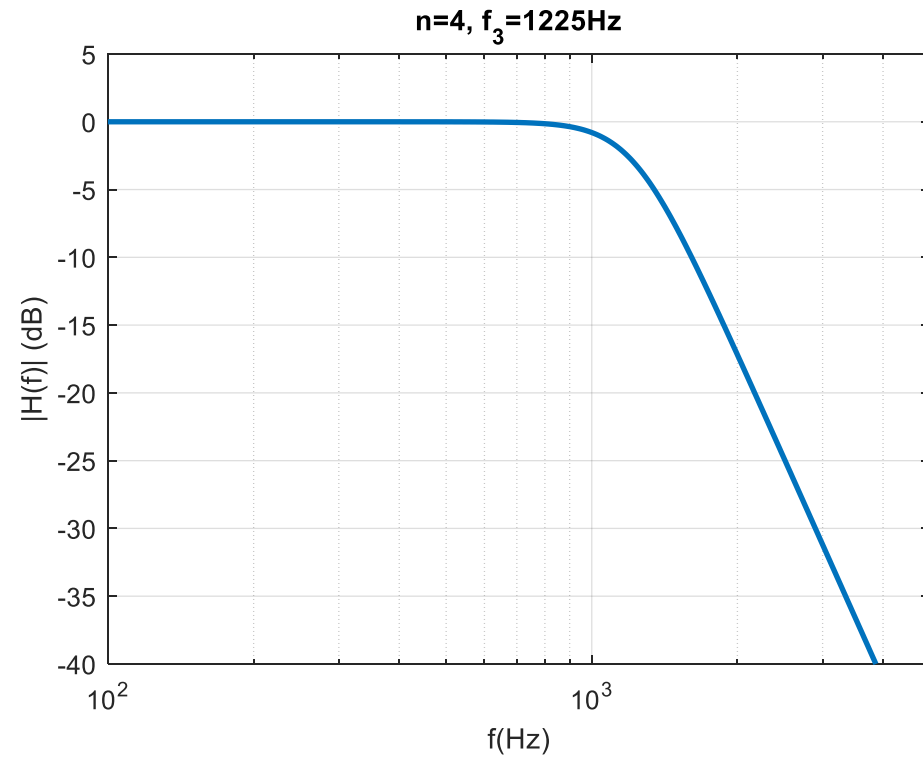
Example



Example 16.1: Design a Butterworth LPF to meet the following specifications:

Passband: $|f| < 1\text{kHz}$, $|H(f)|^2 > -1\text{dB}$,

Stopband: $|f| > 3\text{kHz}$, $|H(f)|^2 < -30\text{dB}$.



An Alternative Design Procedure



The following is an alternative design procedure that will allow us to generalize to other classes of filters.

- Start from the Butterworth magnitude response

$$|H(f)|^2 = \frac{1}{1 + \left(\frac{f}{f_3}\right)^{2n}},$$

- Rewrite this response in terms of the passband edge frequency, f_1 ,

$$|H(f)|^2 = \frac{1}{1 + \varepsilon \left(\frac{f}{f_1}\right)^{2n}}.$$

- The parameter ε is chosen so that $|H(f_1)|^2 = \alpha_1$,

$$|H(f_1)|^2 = \frac{1}{1 + \varepsilon} = \alpha_1 \rightarrow \varepsilon = \frac{1}{\alpha_1} - 1.$$

- The order of the filter is then chosen to meet the stopband specification

$$|H(f_2)|^2 = \frac{1}{1 + \varepsilon \left(\frac{f_2}{f_1}\right)^{2n}} \leq \alpha_2 \rightarrow n \geq \frac{1}{2} \log \left(\frac{\frac{1}{\alpha_2} - 1}{\varepsilon} \right) / \log \left(\frac{f_2}{f_1} \right).$$

- This is the same equation as before.

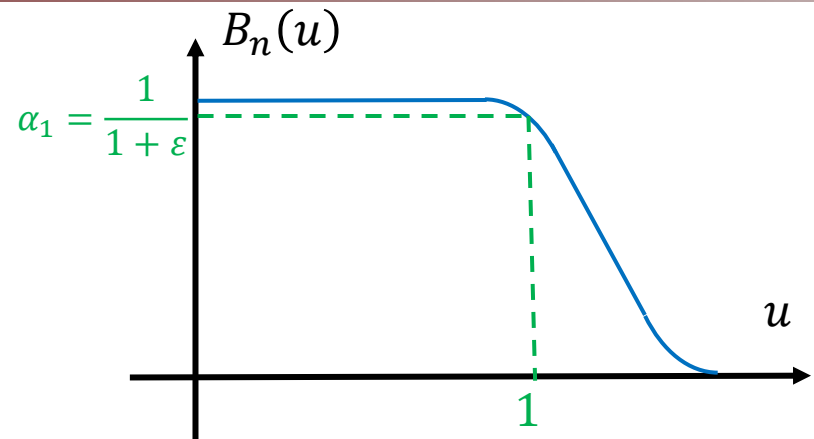
A Generalization



In order to develop some alternative classes of filters, rewrite the Butterworth response as

$$B_n(u) = \frac{1}{1 + \varepsilon u^{2n}} = \frac{1}{1 + \varepsilon (u^n)^2},$$

where $u = f/f_1$ is a normalized frequency.



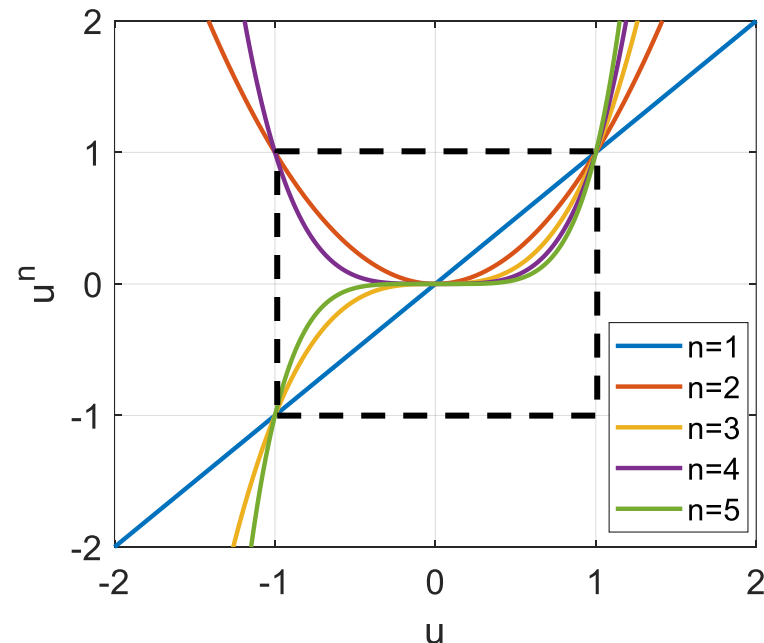
Next, consider a more general class of filters of the form

$$T_n(u) = \frac{1}{1 + \varepsilon (C_n(u))^2}.$$

where $C_n(u)$ can be any n^{th} order polynomial in the variable u . In order for the filter to meet the passband specification, we need

$$T_n(u) \geq \alpha_1 = \frac{1}{1+\varepsilon} \rightarrow |C_n(u)| \leq 1 \text{ for all } |u| \leq 1.$$

Then for $|u| > 1$, we want $C_n(u)$ to grow as rapidly as possible to create a sharp cutoff.



A First Example



Example 16.2: 2nd order filter

We get a greater slope at $u = \pm 1$ with a 2nd order polynomial by choosing $C_2(u)$ so that it is an even polynomial that passes through (1,1) and has a local minimum value of -1 at $u = 0$.

$$\rightarrow C_2(u) = 2u^2 - 1.$$

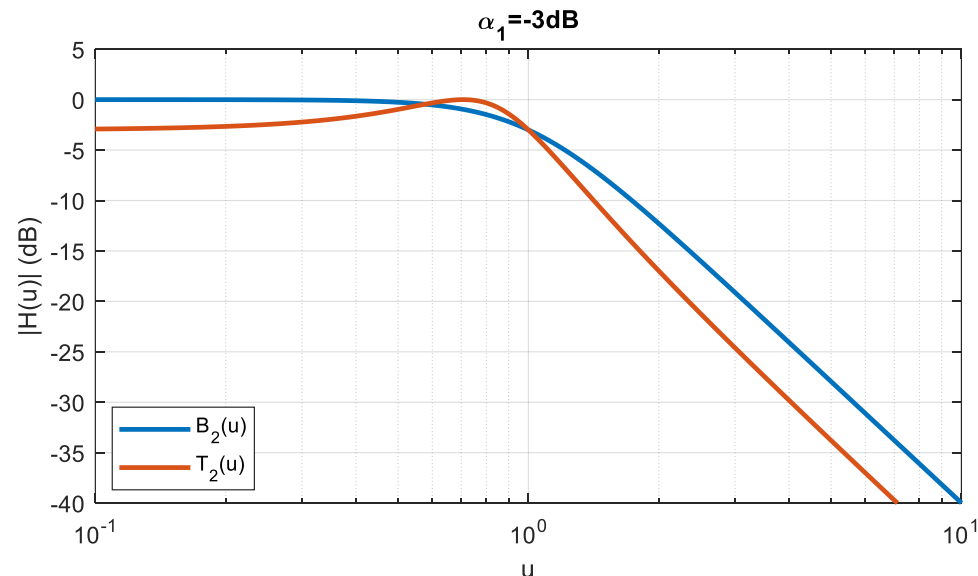
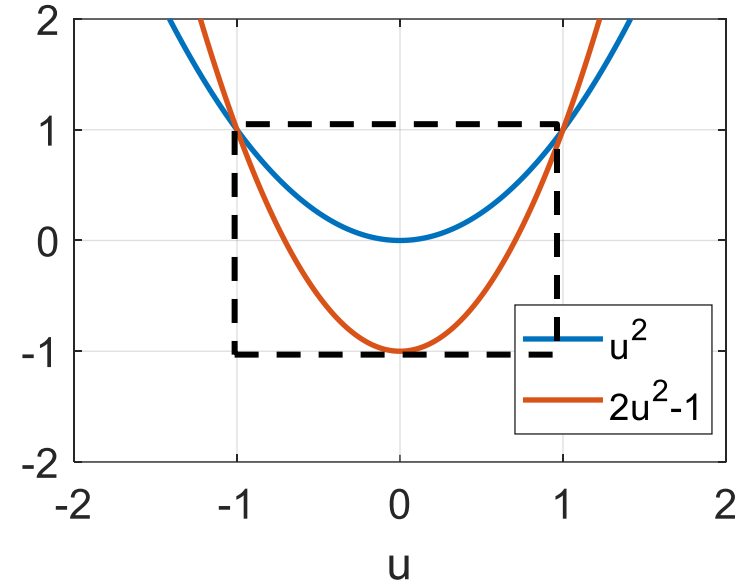
$$T_2(u) = \frac{1}{1 + \varepsilon(2u^2 - 1)^2}$$

$$= \frac{1}{1 + \varepsilon[1 - 4u + 4u^2]}$$

$$= \frac{1}{1 + \varepsilon - 4\varepsilon u + 4\varepsilon u^2}.$$

$$|H(f)|^2 = T_2(u) \Big|_{u=f/f_1}$$

$$= \frac{1}{1 + \varepsilon - 4\varepsilon(f/f_1) + 4\varepsilon(f/f_1)^2}$$



More of the same



Example 16.3: 3rd order filter

$$B_3(u) = \frac{1}{1 + \varepsilon[u^3]^2}.$$

Choose $C_3(u)$ to be an odd polynomial that passes through (1,1) and has a local minimum value of -1 somewhere in the range $0 < u < 1$.

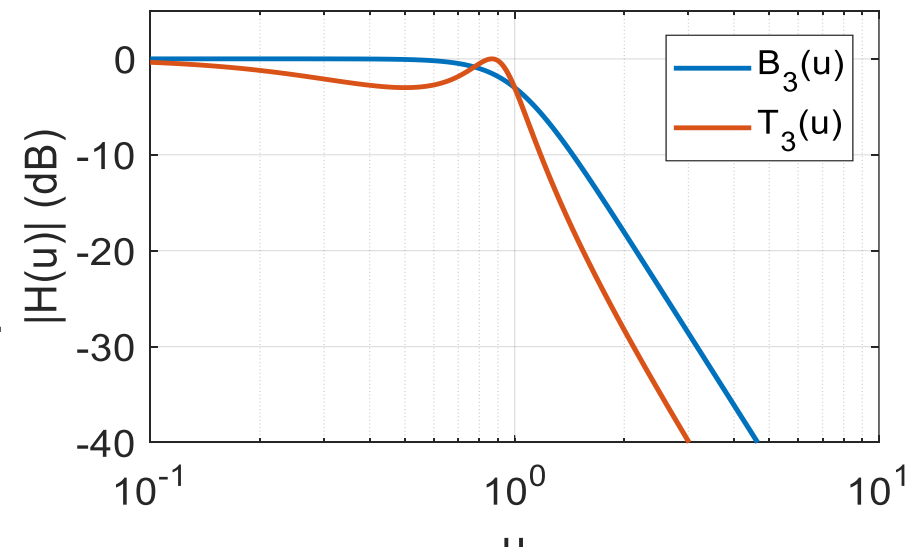
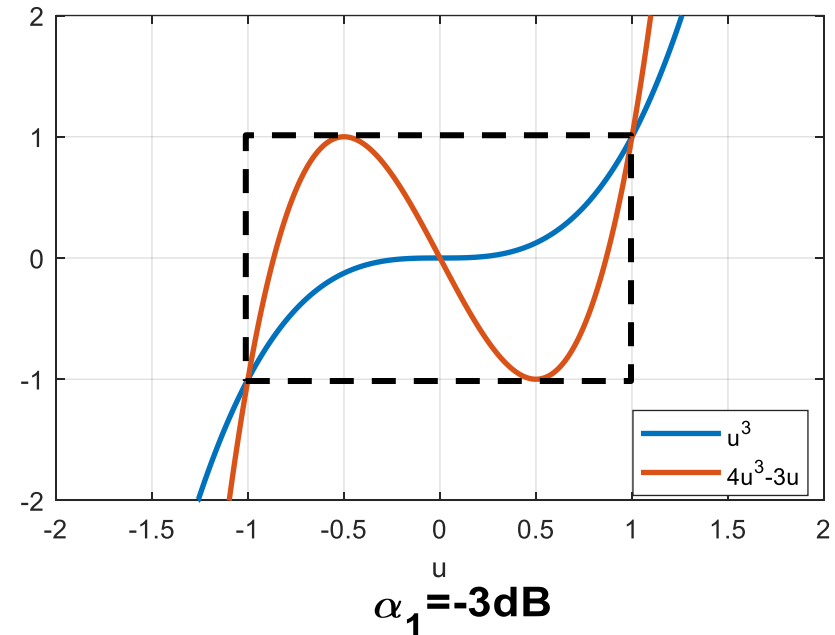
$$\rightarrow C_3(u) = 4u^3 - 3u.$$

$$T_3(u) = \frac{1}{1 + \varepsilon(4u^3 - 3u)^2}$$

$$= \frac{1}{1 + \varepsilon[9u^2 - 24u^4 + 16u^6]}.$$

$$|H(f)|^2 = T_2(u) \Big|_{u=f/f_1}$$

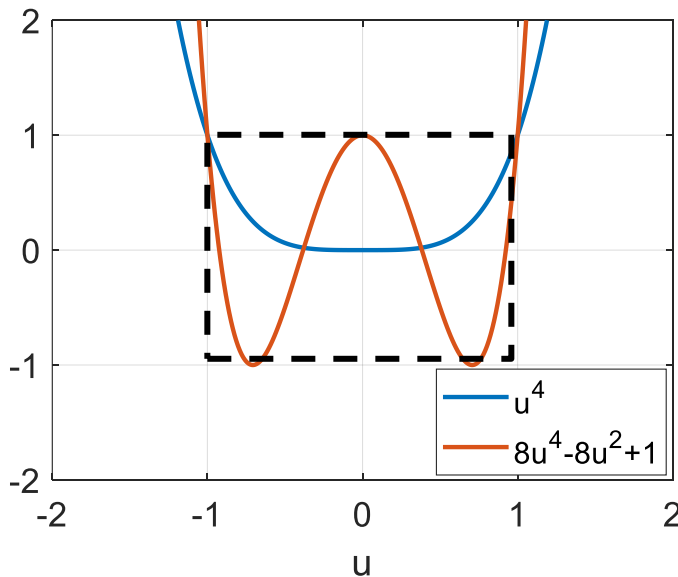
$$= \frac{1}{1 + \varepsilon[9(f/f_1)^2 - 24(f/f_1)^4 + 16(f/f_1)^6]}.$$



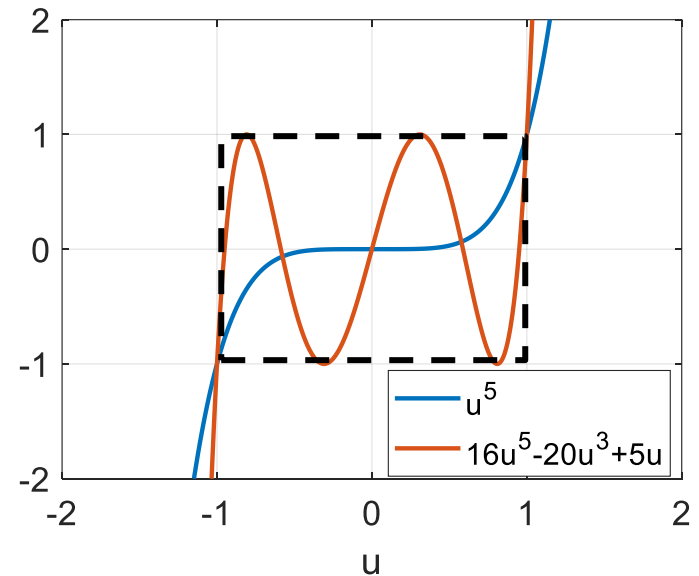
Chebyshev Polynomials



Following similar reasoning, we can get higher order polynomials



$$C_4(u) = 8u^4 - 8u^2 + 1$$



$$C_5(u) = 16u^5 - 20u^3 + 5u$$

The polynomials obtained are known as Chebyshev polynomials and have the “equiripple” property. They can be found recursively according to

$$C_0(u) = 1,$$

$$C_1(u) = u,$$

$$C_n(u) = 2uC_{n-1}(u) - C_{n-2}(u), \quad n = 2, 3, 4, \dots$$

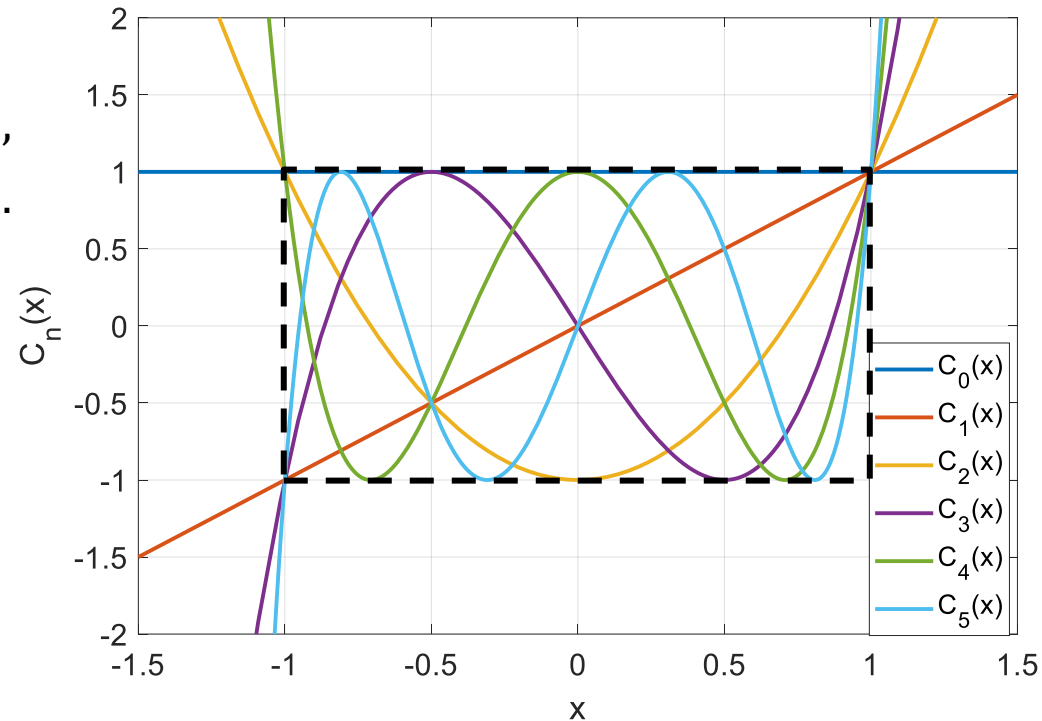
Chebyshev Polynomials



Chebyshev polynomials can be expressed as

$$C_n(u) = \begin{cases} \cos(n \cos^{-1}(u)), & |u| \leq 1, \\ \cosh(n \cosh^{-1}(u)), & |u| \geq 1. \end{cases}$$

n	$C_n(u)$
0	1
1	u
2	$2u^2 - 1$
3	$4u^3 - 3u$
4	$8u^4 - 8u^2 + 1$
5	$16u^5 - 20u^3 + 5u$
6	$32u^6 - 48u^4 + 18u^2 - 1$
7	$64u^7 - 112u^5 + 56u^3 - 7u$
8	$128u^8 - 256u^6 + 160u^4 - 32u^2 + 1$
9	$256u^9 - 576u^7 + 432u^5 - 120u^3 + 9u$



Chebyshev polynomials are solutions to the differential equation:

$$(1 - x^2)y''(x) - xy'(x) + n^2y(x) = 0.$$

They also satisfy the “orthogonality” property

$$\int_{-1}^1 \frac{C_n(x)C_m(x)}{\sqrt{1-x^2}} dx = 0 \text{ for } n \neq m.$$

Lowpass Chebyshev Filter Design

For the n^{th} order Chebyshev filter, the magnitude response is

$$|H(f)|^2 = \frac{1}{1 + \varepsilon [C_n(f/f_1)]^2}.$$

To meet the passband specification, we choose

$$|H(f_1)|^2 = \frac{1}{1 + \varepsilon [C_n(f_1/f_1)]^2} = \frac{1}{1 + \varepsilon} \geq \alpha_1 \rightarrow \varepsilon \leq \frac{1}{\alpha_1} - 1.$$

Similarly, to meet the stopband specification we must satisfy

$$|H(f_2)|^2 = \frac{1}{1 + \varepsilon [C_n(f_2/f_1)]^2} \leq \alpha_2 \rightarrow \varepsilon \geq \frac{\frac{1}{\alpha_2} - 1}{[C_n(f_2/f_1)]^2}$$

To meet both the passband and stopband specifications, the parameter ε must be chosen within the range

$$\frac{\frac{1}{\alpha_2} - 1}{[C_n(f_2/f_1)]^2} \leq \varepsilon \leq \frac{1}{\alpha_1} - 1$$

Lowpass Chebyshev Filter Design

Choosing within the range specified on the previous slide is only achievable if the lower bound is smaller than the upper bound. Therefore, the order of the filter must be chosen to satisfy

$$\frac{\frac{1}{\alpha_2} - 1}{[C_n(f_2/f_1)]^2} \leq \frac{1}{\alpha_1} - 1 \rightarrow C_n(f_2/f_1) \geq \sqrt{\frac{\frac{1}{\alpha_2} - 1}{\frac{1}{\alpha_1} - 1}}$$

In order to solve this inequality for n , use the trig function formulation of the Chebyshev polynomial

$$C_n(u) = \cosh(n \cosh^{-1}(u)) \geq y \rightarrow n \geq \frac{\cosh^{-1}(y)}{\cosh^{-1}(u)}.$$

Thus, the filter order must be chosen to satisfy

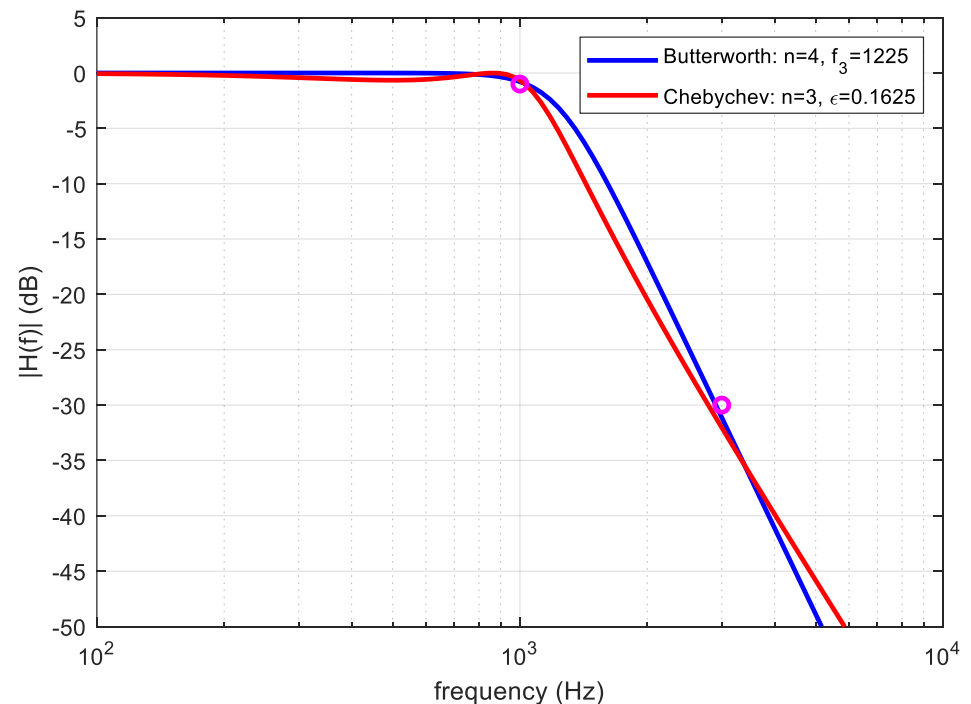
$$n \geq \cosh^{-1} \left(\sqrt{\left(\frac{1}{\alpha_2} - 1\right) / \left(\frac{1}{\alpha_1} - 1\right)} \right) / \cosh^{-1} \left(\frac{f_2}{f_1} \right).$$

Example



Example 16.1 (revisited) : Design a Chebyshev LPF to meet the following specifications:

Passband: $|f| < 1\text{kHz}$, $|H(f)|^2 > -1\text{dB}$,
Stopband: $|f| > 3\text{kHz}$, $|H(f)|^2 < -30\text{dB}$.





Poles of Chebyshev LPFs



The pole locations for a Chebyshev filter can be found in a manner similar to what was done for the Butterworth filter (although the details are a bit more complicated).

First, since the magnitude response is of the form

$$|H(u)|^2 = \frac{1}{1 + \varepsilon[C_n(u)]^2},$$

There are potential pole locations (in the normalized frequency domain) at

$$1 + \varepsilon[C_n(u)]^2 = 0 \rightarrow C_n(u) = \pm \sqrt{-\frac{1}{\varepsilon}} = \pm j \sqrt{\frac{1}{\varepsilon}}.$$

$$\cosh(ncosh^{-1}(u)) = \pm j\sqrt{1/\varepsilon}$$

$$u = \cosh\left(\frac{1}{n} \cosh^{-1}(\pm j\sqrt{1/\varepsilon})\right).$$

After a bunch of algebra, it can be shown that the solutions to this equation are

$$u_k = \cos\left(\frac{(2k-1)\pi}{2n}\right) \cosh(z) + j \sin\left(\frac{(2k-1)\pi}{2n}\right) \sinh(z), \quad k = 1, 2, 3, \dots, 2n,$$

$$z = \frac{1}{n} \ln\left(\sqrt{1 + \frac{1}{\varepsilon}} + \sqrt{\frac{1}{\varepsilon}}\right).$$

Poles of Chebyshev LPFs



Writing $f = uf_1$ and $s = j2\pi f$, these potential pole locations can then be written in the s-domain as

$$s_k = \omega_1 \sin\left(\frac{(2k-1)\pi}{2n}\right) \sinh(z) + j\omega_1 \cos\left(\frac{(2k-1)\pi}{2n}\right) \cosh(z),$$

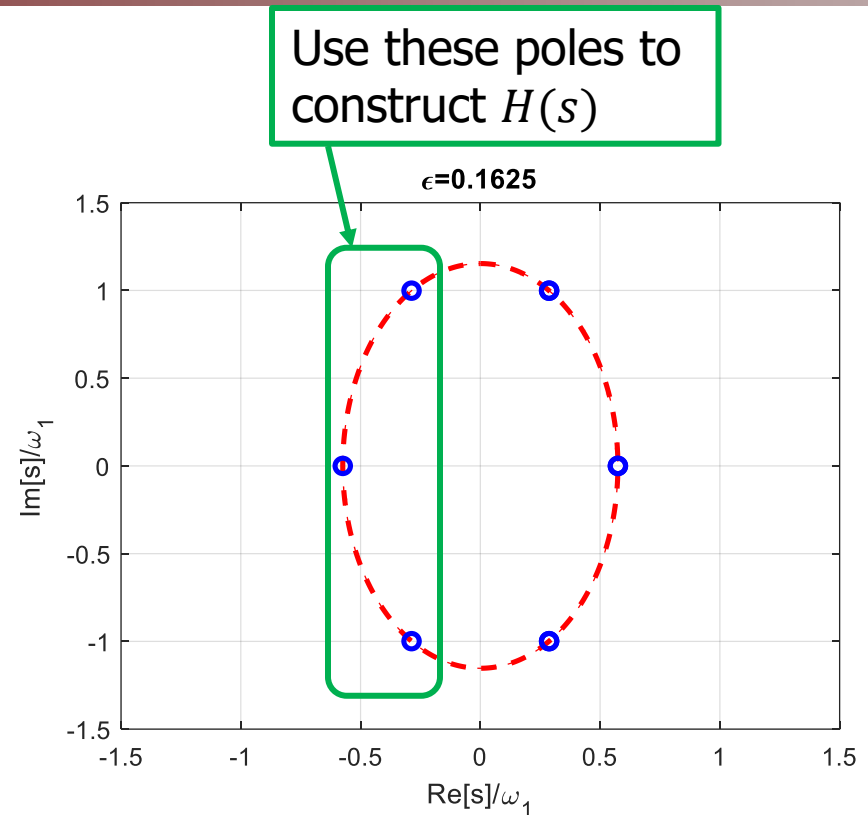
$$k = 1, 2, 3, \dots, 2n,$$

$$z = \frac{1}{n} \ln\left(\sqrt{1 + \frac{1}{\varepsilon}} + \sqrt{\frac{1}{\varepsilon}}\right)$$

$$= \frac{1}{n} \sinh^{-1}(\sqrt{1/\varepsilon})$$

It turns out that these locations fall on an ellipse in the s-plane, whose foci are at $s = \pm j\omega_1$.

Of these $2n$ potential pole locations, choose the n locations in the LHP to construct $H(s)$.



From the previous example:

$$f_1 = 1 \text{ kHz}$$

$$\varepsilon = \frac{13}{80} = 0.1625,$$

$$z = \frac{1}{n} \sinh^{-1}(\sqrt{1/\varepsilon}) = 0.5467,$$

$$s_k = [-3608.6, -1804.3 \pm j6275.0]$$